

AI Agentic Operations System

Comprehensive Technical & Non-Technical Architecture, Operations, Resilience & Monitoring Manual

Version: 1.0.0 (Production Blueprint)	Classification: Enterprise Confidential	System Core: LangGraph + FastAPI + React
Author: Condigence Engineering	Target Audience: Business Owners & Developers	Status: Active & Verified

1. Non-Technical Executive Overview

The **Condigence AI Agentic Operations System** is an autonomous multi-agent operational platform designed to automate and orchestrate core enterprise workflows—including corporate compliance, financial drafting, client communications, HR operations, project tracking, and executive analytics—while enforcing **mandatory Human-in-the-Loop (HITL) authorization gates**.

Key Takeaway for Business Owners & Executives:

AI agents do not act uncontrollably. They research, calculate, structure, and prepare draft invoices, statutory filings, and emails. However, no financial disbursement, client email, or legal document is released without explicit human partner sign-off in the HITL Approval Queue.

How the Business Workflow Operates (Step-by-Step)

- **Step 1: Strategic Instruction:** An executive submits an operational directive in plain language (e.g. *'Draft invoice for Acme Corp for INR 50,000 and notify them via email'*).
- **Step 2: AI CEO Strategic Planning:** The virtual CEO analyzes the instruction, breaks it down into required specialist tasks, and constructs a tactical execution plan.
- **Step 3: Autonomous Specialist Execution:** The AI Supervisor assigns worker agents in exact order (e.g. Finance Agent calculates taxes and prepares the invoice → Comms Agent crafts the client email).
- **Step 4: Quality & Safety Evaluation:** Automated safety evaluators verify calculation validity (e.g. ensuring positive amounts, correct GST rates, and standard statutory formats).
- **Step 5: Human Partner Authorization (HITL):** Generated artifacts appear in the Executive Portal with interactive previews. The partner clicks *Approve & Release* or *Reject*.
- **Step 6: Immutable Audit Trail:** Every single action, reasoning step, and partner approval is permanently recorded with cryptographic timestamps.

Business Benefits Matrix

Operational Pillar	Traditional Manual Workflow	Condigence Agentic System
Finance & Billing	Manual Zoho entry, manual 18% GST calculation, risk of formula errors.	Instant structured drafting, automated statutory GST engine, 1-click partner sign-off.
MCA/ROC Compliance	Spreadsheet tracking, missed Form 8/Form 11 deadlines, penalty exposure.	Autonomous regulatory monitoring, proactive statutory filing dossier generation.
Client Comms	Delayed email drafting, inconsistent tone, scattered email threads.	Context-aware professional drafts, omnichannel alignment (Gmail/WhatsApp).
Payroll & HR	Manual leave reconciliation, complex TDS deduction math.	Automated department breakdown, gross-to-net disbursement calculation.

2. Technical Architecture & Component Specification

The system implements a decoupled, high-concurrency client-server architecture powered by FastAPI and LangGraph with stateful event reduction.

Full Technology Stack Matrix

Layer	Technologies Used	Architectural Purpose & Responsibility
Frontend Client	React 18, Vite 5, Lucide-React, CSS3 Design Tokens	Executive Single Page Application (SPA), live directive dashboard, HITL queue, and visual trace inspector.
API Gateway	FastAPI 0.110+, Uvicorn, Pydantic v2, WebSockets	Asynchronous non-blocking REST gateway, lifespan connection managers, and WebSocket event emitters.
Agent Orchestration	LangGraph 1.2+, LangChain Core, StateGraph	Directed Acyclic Graph (DAG) state machine, multi-agent reducer state management, and supervisor routing.
AI / LLM Layer	Groq (LLaMA-3.3 70B), Gemini 2.0, OpenAI GPT-4o	Multi-tier LLM inference with ultra-fast execution, structured schema decoding, and fallback recovery.
Persistence & Telemetry	MongoDB Atlas, Motor, Redis 8, Apache Kafka	Immutable audit event streams, task state persistence, token bucket rate-limiting, and distributed caching.

Specialized Multi-Agent Fleet Hierarchy

The system partitions enterprise intelligence into 7 distinct roles:

Agent Name	Role & Domain	Input Directive	Generated Output / Artifact
AI_CEO	Strategic visionary & planner	Executive strategic goal	strategic_plan: Identified agents, decomposed sub-tasks, entities.
FinanceAgent	Billing, GST, Zoho Books	Billing directives	invoice_draft: Base amount, 18% GST, invoice number, line items.
CommsAgent	Gmail, WhatsApp, CRM	Message/Notification requests	comms_draft: Recipient, subject, contextual body copy.
ComplianceAgent	MCA, ROC, LLP legalities	Statutory filing requests	compliance_filing: Form 8/11 checklist, authority deadlines.
HRAgent	Payroll, Leaves, Onboarding	Disbursement/HR goals	payroll_draft: Gross disbursement, TDS, net pay, headcounts.
ProjectAgent	Notion board synchronization	Sprint/Milestone queries	project_status: Active sprints, blocked milestones, highlights.
AnalyticsAgent	P&L;, Cash flow forecasting	Financial metric requests	financial_report: Gross revenue, OpEx, margin, cash health.

3. LLM Orchestration, Fallback Strategy & Error Handling

Enterprise reliability requires that transient LLM outages, API throttling, or malformed outputs never crash the application.

Multi-Tiered LLM Fallback Architecture

The system implements a 4-tier graceful degradation strategy in `app/core/llm.py`:

- **Tier 1 (High-Performance Primary):** Groq LLaMA-3.3-70B-Versatile (sub-second latency, structured extraction).
- **Tier 2 (High-Context Secondary):** Google Gemini 2.0 Flash via OpenAI compatibility layer.
- **Tier 3 (Enterprise Tertiary):** OpenAI GPT-4o / GPT-4o-mini.
- **Tier 4 (Deterministic Heuristic Fallback):** Safe rule-based algorithmic drafting ensuring 100% uptime even during complete cloud outages.

Detailed Error Handling Architecture

Failure Vector	Root Cause / Scenario	Automated Mitigation & Recovery Mechanism
LLM Markdown & Token Contamination	Reasoning models outputting <think> tags or conversational text before JSON.	<code>extract_json_from_llm()</code> strips reasoning tags via RegEx, inspects code fences, and uses outermost bracket matching.
Worker Route Deadlock	Worker finishes but CEO had planned multiple downstream agents.	<code>route_from_worker()</code> dynamically compares <code>strategic_plan</code> against <code>completed_steps</code> and auto-chains next agent.
Safety Rule Violation	Worker outputs negative invoice total or truncated email body.	<code>SupervisorEvaluator.validate_step()</code> inspects math and payload integrity before registering for HITL review.
Backend Network Disconnection	FastAPI process stopped or dev proxy connection refused.	React UI displays real-time Backend Offline banner with 1-click Reconnect and surfaces exact HTTP status codes.

4. Output Monitoring, Observability & Auditing

Output verification occurs at three synchronized layers:

- **Layer 1 — Automated Supervisor Evaluation:** Every worker artifact passes through SupervisorEvaluator to ensure domain constraints (positive totals, valid tax rates, mandatory disclaimer copy).
- **Layer 2 — Immutable Audit Logging (AuditService):** Every action is logged with actor type (HUMAN / AGENT), exact payload, timestamp, and status. Audit events are streamed to Apache Kafka (condigence.audit.events) and stored in MongoDB Atlas.
- **Layer 3 — Real-Time Executive Trace Inspector:** The web portal provides live step-by-step visual execution flow, artifact rendering cards, and raw JSON payload inspection.

5. Future Upgrade & Extension Roadmap

- **1. Live External API Integrations:** Connect Zoho Books OAuth2 for direct invoice push, Gmail API / WhatsApp Cloud API for client dispatch, and Notion API for two-way task sync upon human approval.
- **2. Database Persistence Migration:** Transition in-memory task dictionary to distributed MongoDB collections with automated indexing.
- **3. Role-Based Access Control (RBAC):** Introduce granular partner authorization (e.g. Partner A signs financial invoices > INR 1,00,000; Partner B signs MCA filings).
- **4. Voice Directive Input:** Integrate Whisper speech-to-text for hands-free executive directive dispatching.
- **5. Vector Memory & RAG Knowledgebase:** Connect Pinecone/Qdrant vector stores with LLP historical agreements and past MCA resolutions.